



```
In [1]: #Import Libraries
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: #Load dataset
import pandas as pd
titanic = pd.read_csv('/content/train_and_test2.csv')
```

```
In [5]: # View first 5 rows
print(titanic.head())
```

	Passengerid	Age	Fare	Sex	sibsp	zero	zero.1	zero.2	zero.3	\
0	1	22.0	7.2500	0	1	0	0	0	0	
1	2	38.0	71.2833	1	1	0	0	0	0	
2	3	26.0	7.9250	1	0	0	0	0	0	
3	4	35.0	53.1000	1	1	0	0	0	0	
4	5	35.0	8.0500	0	0	0	0	0	0	

	zero.4	...	zero.12	zero.13	zero.14	Pclass	zero.15	zero.16	Embarked
\									
0	0	...	0	0	0	3	0	0	2.0
1	0	...	0	0	0	1	0	0	0.0
2	0	...	0	0	0	3	0	0	2.0
3	0	...	0	0	0	1	0	0	2.0
4	0	...	0	0	0	3	0	0	2.0

	zero.17	zero.18	Survived
0	0	0	0
1	0	0	1
2	0	0	1
3	0	0	1
4	0	0	0

[5 rows x 28 columns]

```
In [6]: # Basic info
print(titanic.info())
```

```

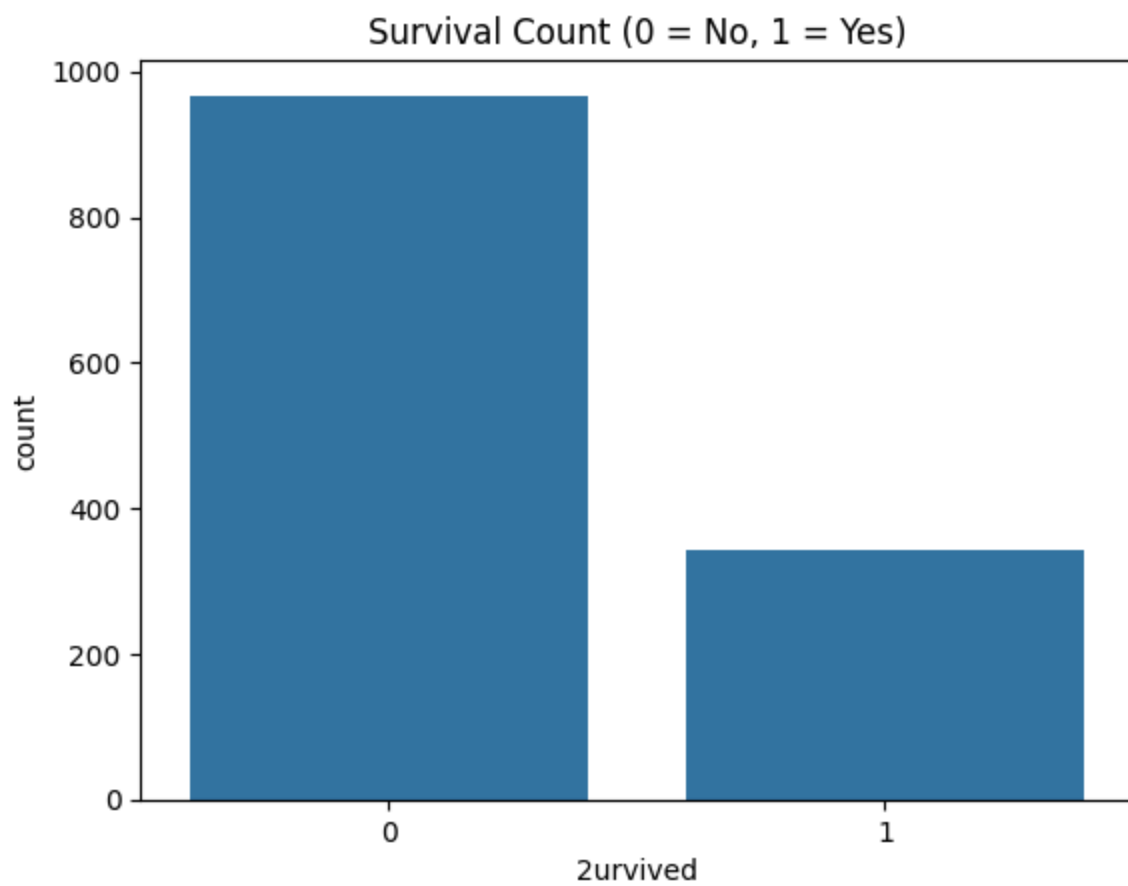
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1309 entries, 0 to 1308
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Passengerid           1309 non-null   int64
1   Age                   1309 non-null   float64
2   Fare                  1309 non-null   float64
3   Sex                   1309 non-null   int64
4   sibsp                 1309 non-null   int64
5   zero                  1309 non-null   int64
6   zero.1                1309 non-null   int64
7   zero.2                1309 non-null   int64
8   zero.3                1309 non-null   int64
9   zero.4                1309 non-null   int64
10  zero.5                1309 non-null   int64
11  zero.6                1309 non-null   int64
12  Parch                 1309 non-null   int64
13  zero.7                1309 non-null   int64
14  zero.8                1309 non-null   int64
15  zero.9                1309 non-null   int64
16  zero.10               1309 non-null   int64
17  zero.11               1309 non-null   int64
18  zero.12               1309 non-null   int64
19  zero.13               1309 non-null   int64
20  zero.14               1309 non-null   int64
21  Pclass                1309 non-null   int64
22  zero.15               1309 non-null   int64
23  zero.16               1309 non-null   int64
24  Embarked              1307 non-null   float64
25  zero.17               1309 non-null   int64
26  zero.18               1309 non-null   int64
27  2urvived              1309 non-null   int64
dtypes: float64(3), int64(25)
memory usage: 286.5 KB
None

```

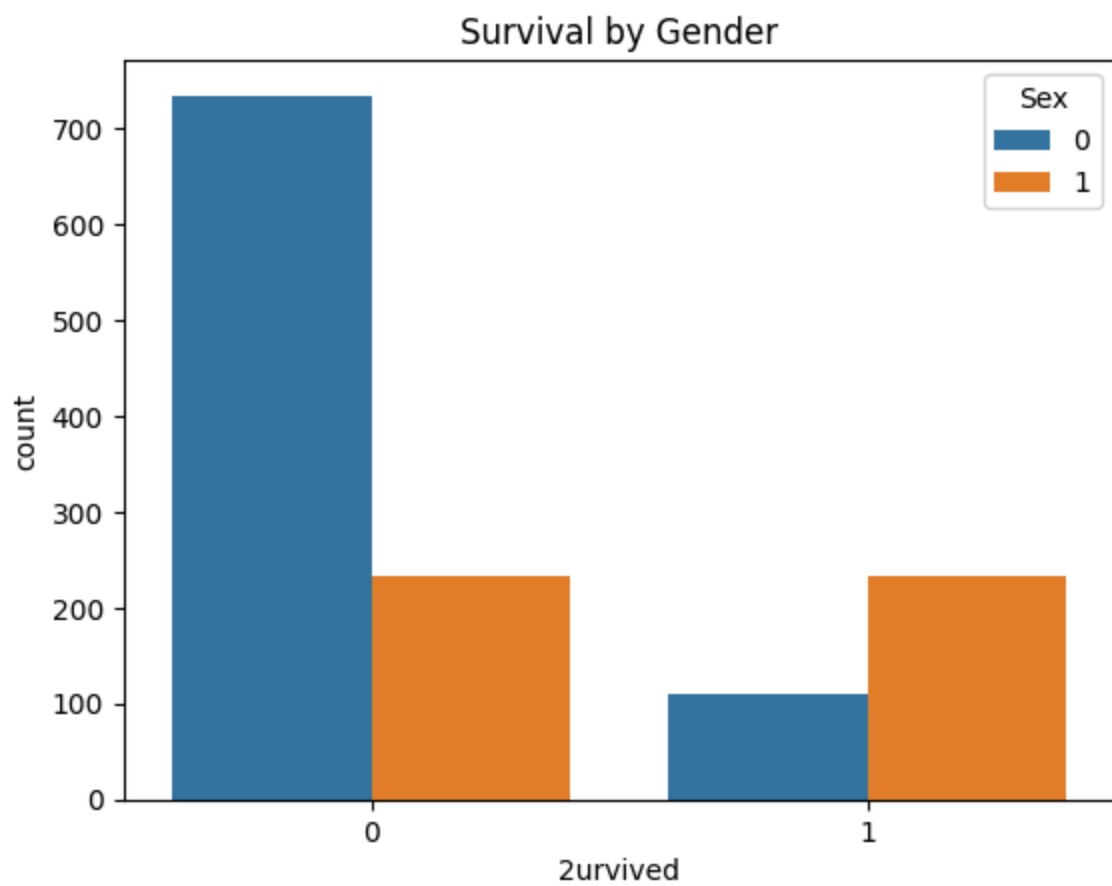
```

In [8]: #Check Survival Count
sns.countplot(x='2urvived', data=titanic)
plt.title("Survival Count (0 = No, 1 = Yes)")
plt.show()

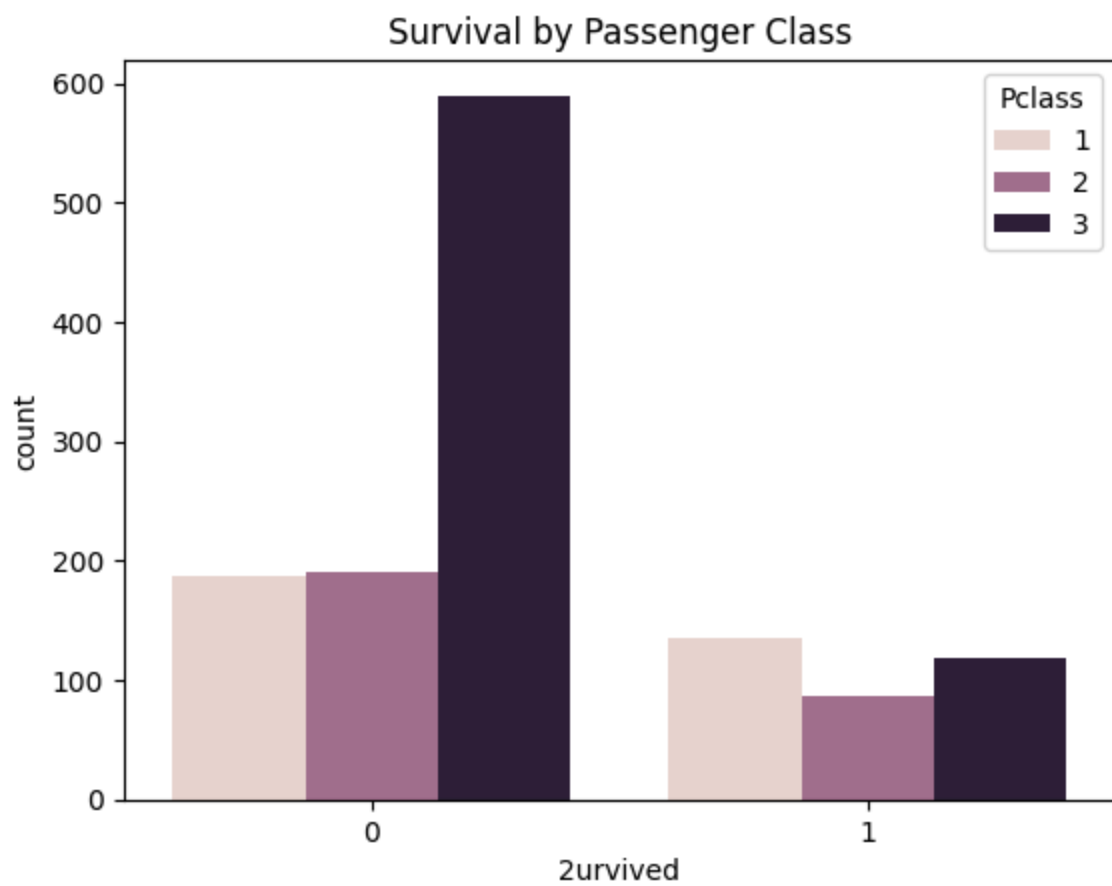
```



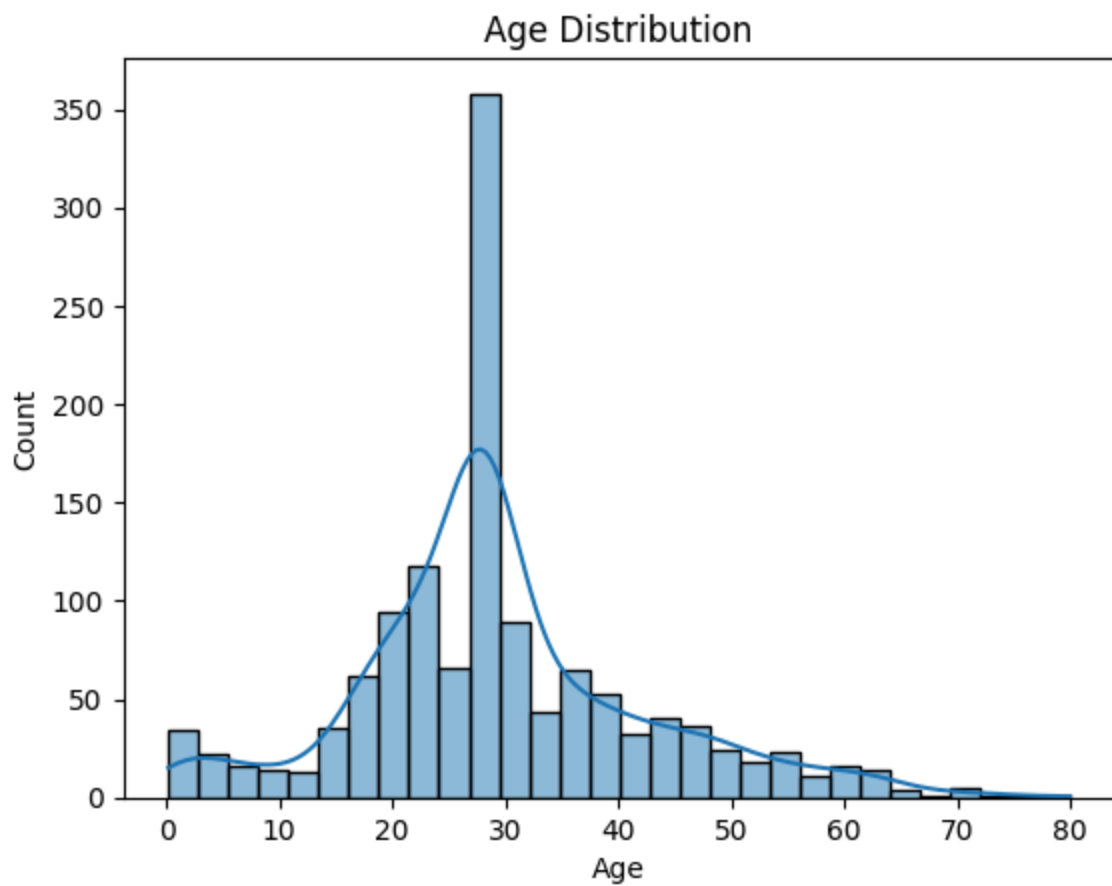
```
In [10]: #Survival Based on Gender
sns.countplot(x='Survived', hue='Sex', data=titanic)
plt.title("Survival by Gender")
plt.show()
```



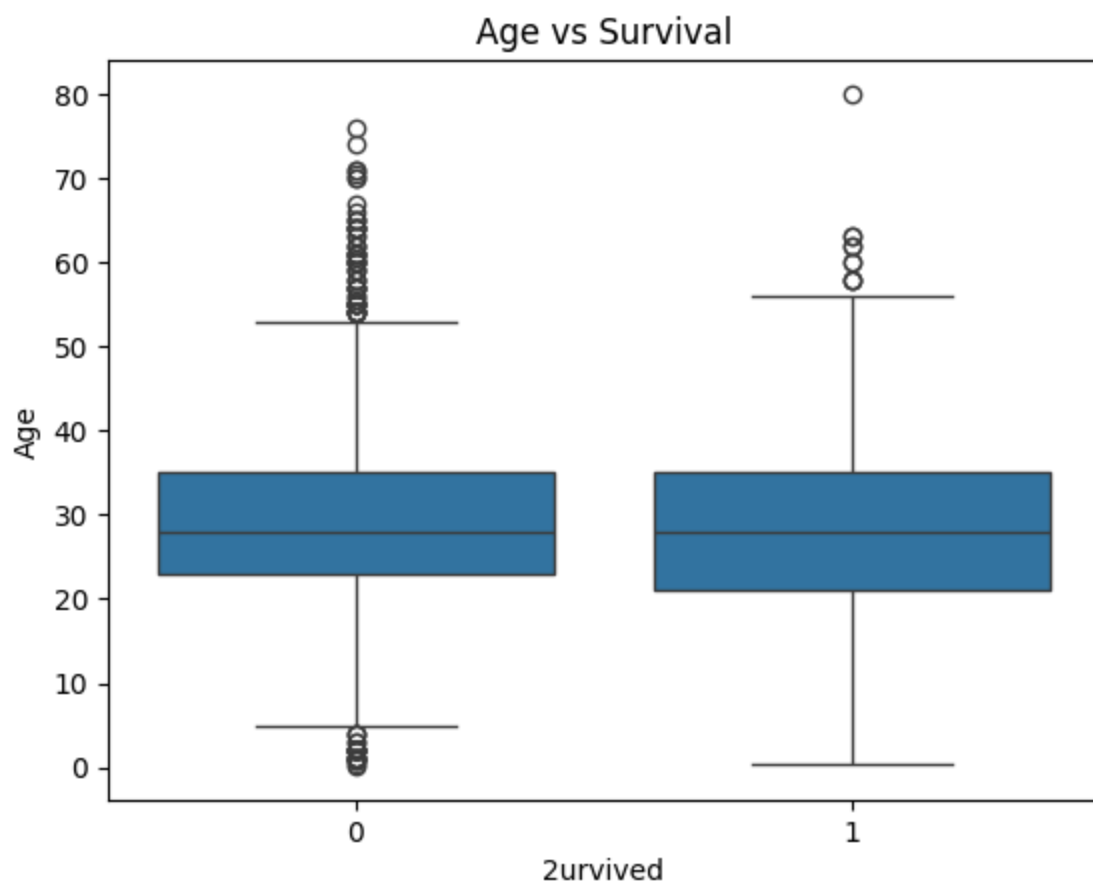
```
In [12]: #Survival Based on Passenger Class
sns.countplot(x='Survived', hue='Pclass', data=titanic)
plt.title("Survival by Passenger Class")
plt.show()
```



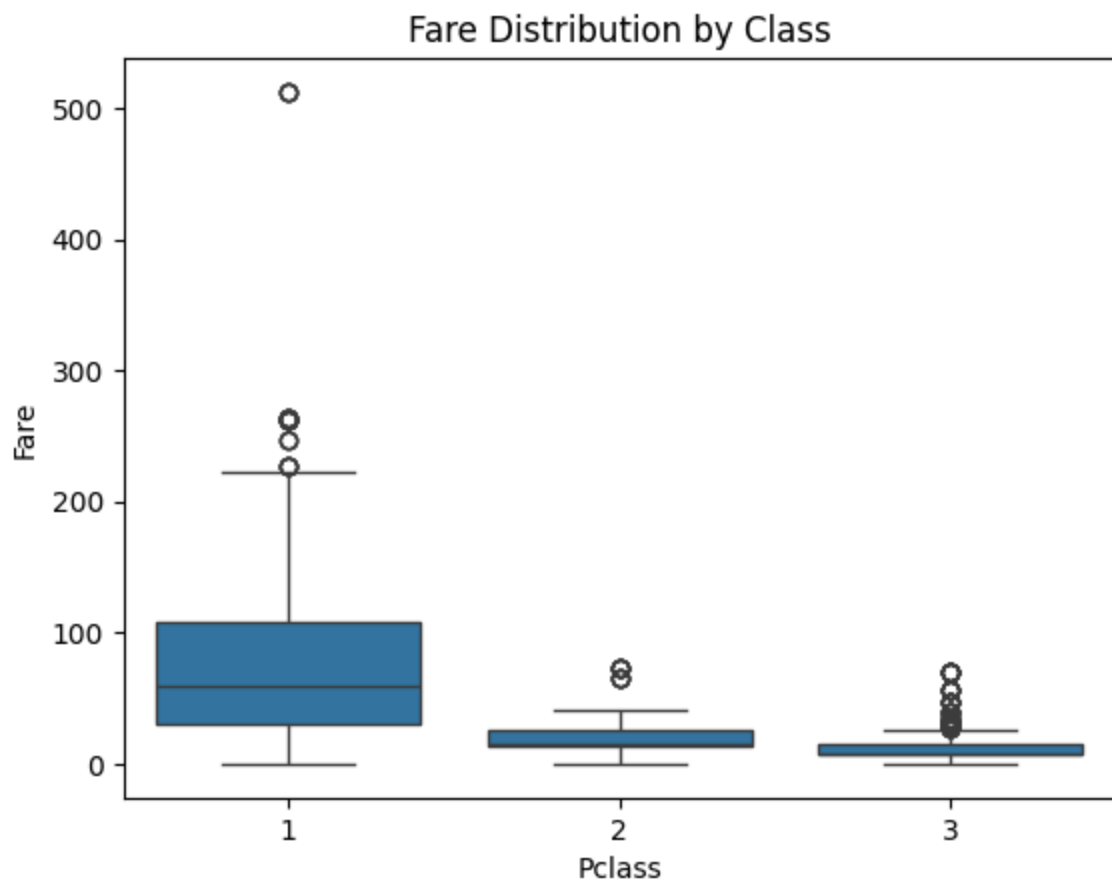
```
In [14]: #Age Distribution
sns.histplot(titanic['Age'].dropna(), bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```



```
In [16]: #Survival vs Age
sns.boxplot(x='Survived', y='Age', data=titanic)
plt.title("Age vs Survival")
plt.show()
```

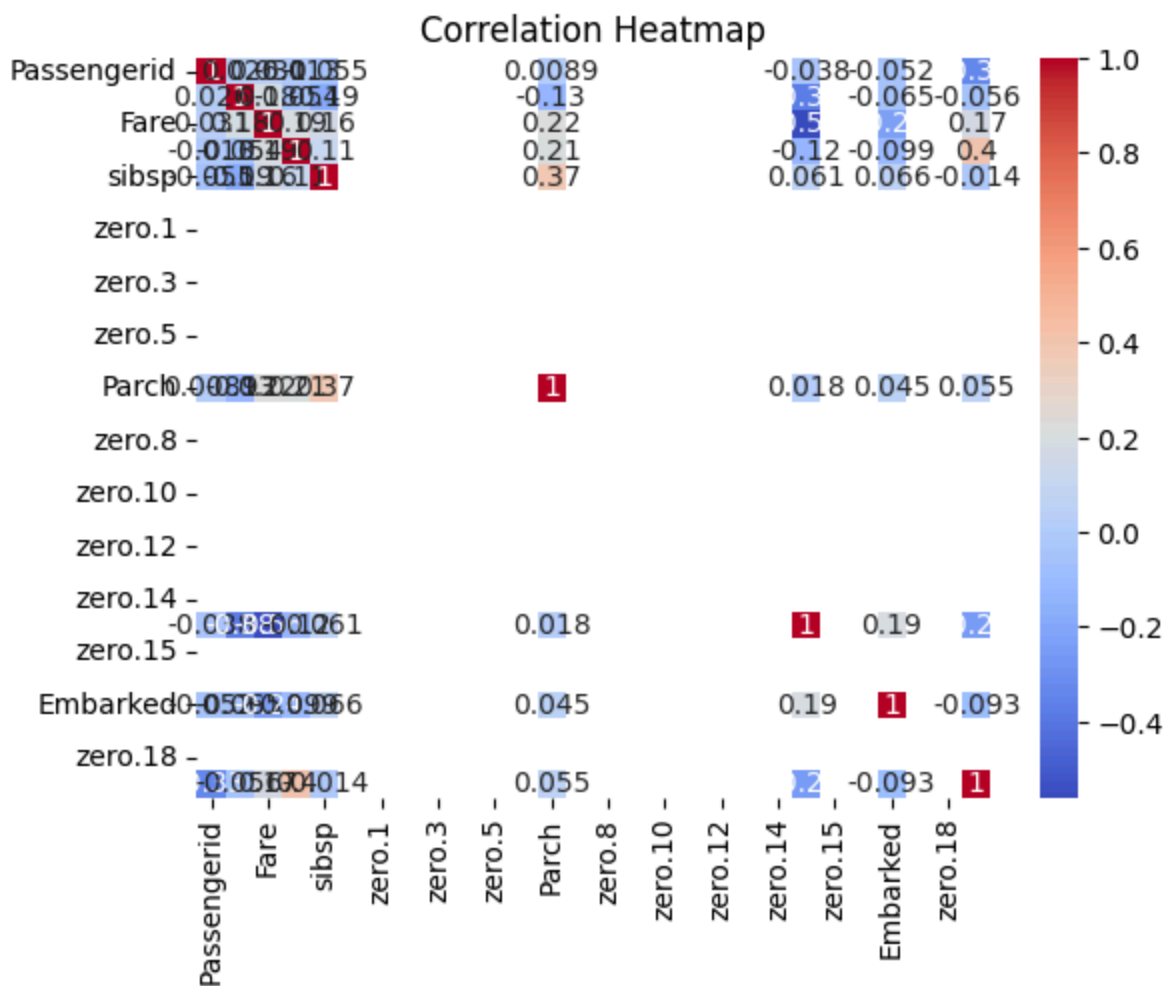


```
In [18]: #Fare Distribution by Class
sns.boxplot(x='Pclass', y='Fare', data=titanic)
plt.title("Fare Distribution by Class")
plt.show()
```



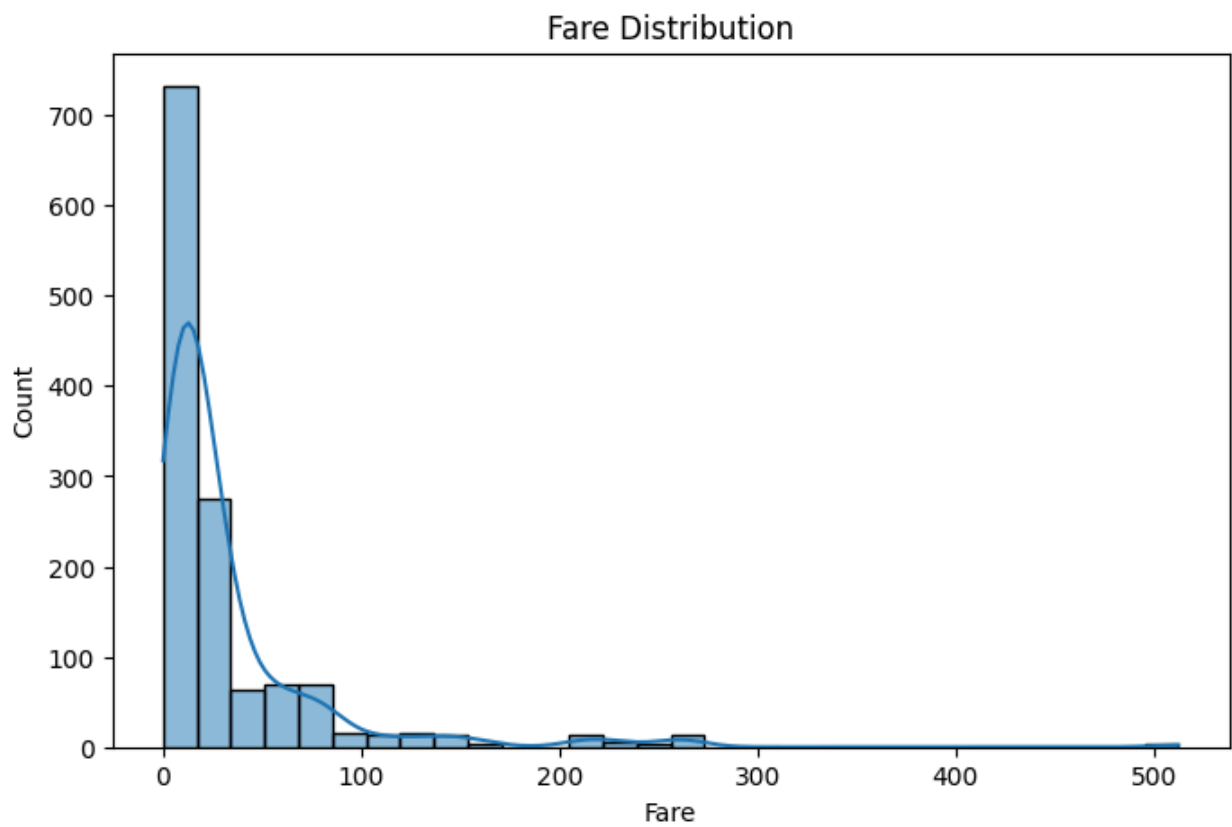
```
In [19]: #Heatmap (Correlation)
corr = titanic.corr(numeric_only=True)
```

```
In [20]: sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

In [22]: # 2. Write a code to check how the price of the ticket (column name: 'fare') is distributed by plotting a histogram.

```
# Create histogram for 'fare'
plt.figure(figsize=(8, 5))
sns.histplot(titanic['Fare'], bins=30, kde=True)
plt.title('Fare Distribution')
plt.xlabel('Fare')
plt.ylabel('Count')
plt.show()
```



```
In [23]: # Add titles and labels
plt.title("Distribution of Ticket Fare")
plt.xlabel("Fare")
plt.ylabel("Count")
```

```
Out[23]: Text(0, 0.5, 'Count')
```

